

Validity Analysis: DRBENCHMARK

Target context: EU Pharmaceutical Regulatory NLP — French

Overall risk: **HIGH**

Deployment Context

Use case: In a pharmaceutical setting, a regulatory specialist uses document management software to ensure compliance in French drug labeling and scientific abstracts by applying a model evaluated on semantic textual similarity and fine-grained named-entity recognition. The system identifies over a dozen types of precise entities, such as chemical compounds and dosage modes, while verifying that safety warnings across different leaflets or patents are semantically consistent. These results determine whether pharmacological documentation meets strict professional standards for official submission or requires manual revision to prevent regulatory non-compliance.

Target population: Regulatory affairs specialists, pharmacologists, and legal experts at pharmaceutical companies or government health agencies.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	2	Significant gaps	high
Input Content 	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology 	1	Serious concern	high
Output Content 	2	Significant gaps	high
Output Form 	4	Minor gaps	high
Average	2.5		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

DrBenchmark is the most comprehensive French biomedical NLU benchmark currently available [Q16, Q83], and parts of its content (QUAERO EMEA, Mantra-GSC fr_emea, DEFT2020 drug-leaflet pairs, PxCorpus posology schema) provide partial signal for the EU pharmaceutical regulatory deployment. However, the deployment's HIGH-priority dimensions — input

ontology, input content, output ontology, and output content — show severe misalignment. The benchmark is calibrated for clinical and research French biomedical NLP, not for EMA/ANSM regulatory affairs workflows: (i) regulatory document genres (SmPCs, CTD modules, EU-RMPs) are absent; (ii) regulatory-specific entity types (INN as distinct from excipient, ATC codes, EMA-templated posology, contraindication qualifiers) are not annotated as distinct classes; (iii) STS scoring is calibrated for general semantic proximity and demonstrably tolerates legally significant micro-differences (2×–60× numeric divergences score 3.0–4.0); (iv) annotators are biomedical NLP / clinical experts, not regulatory affairs specialists, and elicitation explicitly anticipates systematic disagreement on borderline cases. Lower-priority dimensions (input form, output form) are well aligned. Web search confirms several of these gaps are field-level, not just benchmark-level [WEB-11, WEB-12, WEB-16].

ID	Evidence
Q16	“To our knowledge, aside the multilingual benchmark BigBio (Fries et al., 2022) which includes only 4 corpora for French and is initially intended for generative text completion under zero-shot scenario, no large benchmark specialized in the French biomedical field exists.” (p.2)
Q83	“In this paper, we introduced DrBenchmark, the first large language understanding benchmark tailored for the French biomedical domain.” (p.9)
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-12	Bayer et al. 2021 — used pharmacovigilance evaluators for ADE extraction but not regulatory equivalence scoring; closest comparator confirms gap (https://link.springer.com/article/10.1007/s40264-020-00996-3)
WEB-16	IQVIA fact sheet — IDMP requires structured extraction from unstructured SmPC text; ~70% of required attributes reside in unstructured text; this task category is unaddressed by DrBenchmark (https://www.iqvia.com/-/media/iqvia/pdfs/library/fact-sheets/nlp-text-analytics-for-regulatory-affairs.pdf)

Practical Guidance

What This Benchmark Measures

DrBenchmark provides credible measurement of general French biomedical NLU capabilities in clinical and research register: clinical NER (UMLS Semantic Groups), POS, MeSH-axis classification, ICD-10 chapter classification, MCQA over pharmacy exam knowledge, and clinical STS. For the deployment, it offers partial signal on (a) coarse pharmaceutical entity recognition over EMEA leaflet text via QUAERO EMEA and Mantra-GSC fr_emea, (b) French-language STS capability over biomedical sentence pairs including some drug-leaflet content via CLISTER and DEFT2020, and (c) compositional posology recognition via PxCorpus and DEFT2021. Output form (IO-2) is fully aligned, and input form (IF) is mostly aligned.

Construct Depth

Construct probing for the deployment's core needs is shallow. Regulatory-equivalence STS — the deployment's central decision rule — is not measured at all; CLISTER and DEFT2020 measure general semantic proximity, and dataset analysis empirically shows the calibration is incompatible (CLISTER-D12, DEFT2020-D16). Regulatory entity NER is measured only as conflated CHEM/SUBSTANCE classes; INN/ATC/excipient sub-distinctions are unevaluated. Annotation provenance for the most deployment-relevant subsets (QUAERO, Mantra-GSC, CLISTER) is undocumented, so even successful benchmark scores carry unquantified construct-validity risk for borderline regulatory cases.

What Else You Need

To complete the assessment, the deployment team would need: (1) a regulatory-affairs-annotated SmPC/PIL/CTD evaluation set covering INN, ATC, excipient, posology, and contraindication-qualifier entity types — addressing Input Ontology, Input Content, and Output Ontology gaps; (2) a regulatory-equivalence STS rubric and dataset with annotator guidelines that penalize dose-threshold and population-qualifier divergences — addressing Output Ontology calibration and Output Content (annotator alignment); (3) a re-annotation of QUAERO EMEA and Mantra-GSC fr_emea spans by regulatory affairs specialists to quantify systematic disagreement — addressing Output Content; (4) integration of EMA QRD v10.4 [WEB-7, WEB-18] and IDMP [WEB-16, WEB-17] structural extraction targets as evaluation tasks. Given that web search confirms these are field-level gaps [WEB-11, WEB-12], supplementation will require new resource creation, not just resource discovery.

ID	Evidence
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CLISTER-D12	L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — 2× quantitative difference scored 4.0 (very similar)
DEFT2020-D16	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High similarity score despite addition of specific drug names in target — legally significant detail unmarked
WEB-7	EMA QRD-templated documents have formulaic typographic conventions (section numbering, tabular posology) that may stress sentence-splitting at boundaries (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-12	Bayer et al. 2021 — used pharmacovigilance evaluators for ADE extraction but not regulatory equivalence scoring; closest comparator confirms gap (https://link.springer.com/article/10.1007/s40264-020-00996-3)
WEB-16	IQVIA fact sheet — IDMP requires structured extraction from unstructured SmPC text; ~70% of required attributes reside in unstructured text; this task category is unaddressed by DrBenchmark (https://www.iqvia.com/-/media/iqvia/pdfs/library/fact-sheets/nlp-text-analytics-for-regulatory-affairs.pdf)
WEB-17	https://www.linguamatics.com/blog/countdown-idmp-compliance-are-you-ready
WEB-18	EMA QRD v10.4 template PDF — current operational template structure (https://www.ema.europa.eu/en/documents/template-form/qrd-product-information-template-version-104-highlighted_en.pdf)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

DrBenchmark's task taxonomy (POS, NER, MCQA, MCC, MLC, STS) is calibrated for clinical and research NLP rather than EU pharmaceutical regulatory workflow categories [Q81, Q83]. The deployment requires NER and STS over EU regulatory document genres (SmPCs, PILs, CTD modules, EU-RMPs), but the benchmark organizes tasks around general biomedical constructs and includes substantial irrelevant categories: MCQA over pharmacy exam questions [Q39], abstract-level specialty classification [Q38], ICD-10 chapter classification [Q45], and negation/speculation scope [CAS, ESSAI]. The dataset analysis confirms critical task-type mismatches for FrenchMedMCQA, MORFITT, and DiaMED, which evaluate constructs orthogonal to the deployment's NER/STS extraction needs (FRENCHMEDMCQA-D11, MORFITT-D7, DIAMED-D6, DIAMED-D13). No category covers regulatory equivalence scoring as a distinct output decision rule, despite this being a HIGH-priority deployment dimension. Partial overlap exists for NER (QUAERO, MANTRAGSC, DEFT2021, PxCorpus) and STS (CLISTER, DEFT2020), so the benchmark is not entirely orthogonal — but construct underrepresentation for regulatory-specific task types is severe.

ID	Evidence
Q38	"MorFITT (Labrak et al., 2023) is a multi-label dataset annotated with medical specialties. It contains 3,624 biomedical abstracts from PMC Open Access. It has been annotated across 12 medical specialties (more detail in Appendix B.5), for a total of 5,116 annotations." (p.4)
Q39	"FrenchMedMCQA (Labrak et al., 2022) is a Multiple-Choice Question-Answering (MCQA) dataset for biomedical domain. It contains 3,105 questions coming from real exams of the French medical specialization diploma in pharmacy, integrating single and multiple answers. The first task consists of automatically identifying the set of correct answers among the 5 proposed for a given question. The second task consists of identifying the number of answers (between 1 and 5) supposedly correct for a given question." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q81	"Table 8 summarizes the results obtained on average by the considered MLMs when aggregating the tasks into one of the five designated categories: POS, NER, MCQA, MCC (Multi-class classification), MLC (Multi-label classification), or STS tasks." (p.8)
Q83	"In this paper, we introduced DrBenchmark, the first large language understanding benchmark tailored for the French biomedical domain." (p.9)
DIAMED-D6	Monsieur Y. O. âgé de 19 ans a été hospitalisé le 03 février 2012 pour une paraplégie évoluant depuis un mois. — Classic clinical case narrative register; entirely distinct from regulatory document genre
MORFITT-D7	La mise en place par cathétérisme d'une valve aortique (TAVI)... Le registre France-TAVI... va inclure plus de 3 000 cas — Abstract-level classification only; no token-level entity annotations for NER task
FRENCHMED MCQA-D11	Au cours de la leucémie lymphoïde chronique, le myélogramme montre: — MCQA format — tests knowledge retrieval, not NER or STS; task type mismatch
DIAMED-D13	Il s'agit d'un adolescent âgé de 16 ans, sportif (tennis) qui se plaignait depuis plusieurs mois de douleurs mécaniques du coude droit. — Clinical case narrative about orthopedic condition; no pharmaceutical content

Strengths

- Task inventory includes both NER and STS, the two task types the deployment requires [Q1, Q4]

- MANTRAGSC fr_emea and fr_patents subsets provide task categories that overlap genre-wise with the deployment's regulatory and IP workflow components (MANTRAGSC-D1, MANTRAGSC-D7)
- DEFT2020 STS Task 1 contains drug leaflet sentence pairs as a sub-component, structurally analogous to safety-warning equivalence checking (DEFT2020-D1, DEFT2020-D2)

ID	Evidence
Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q4	"These tasks encompass part-of-speech (POS) tagging, named-entity recognition (NER), classification, question-answering (QA), and semantic textual similarity (STS)." (p.1)
MANTRAGSC-D1	Chaque comprimé contient 500 mg de ranolazine. — Drug composition statement with INN annotated CHEM; EMEA register
DEFT2020-D1	En raison de la présence de lactose, ce médicament est contre-indiqué en cas de galactosémie congénitale, de syndrome de malabsorption du glucose et du galactose ou de déficit en lactase. — Contraindication phrasing with excipient (lactose) — directly deployment-relevant regulatory text
DEFT2020-D2	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — Drug safety warning with dose-time threshold — deployment-central entity type
MANTRAGSC-D7	Forme posologique orale selon la revendication 1, dans laquelle ledit antagoniste opioïde libérable est la naltrexone et ledit opioïde libérable est la codéine, dans laquelle le rapport de la naltrexone sur la codéine libérable est de 0,005 : 1 à 0,044 : 1. — French patent claim with INNs and dosage ratio; patent claim register confirmed

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required test case categories for the deployment: (a) NER over EU regulatory document genres (SmPCs, PILs, CTD modules, EU-RMPs) for INN/ATC/excipient/posology/contraindication entity types; (b) regulatory-equivalence STS over EMA-templated text; (c) potentially patent-claim NER for IP workflow component. Per elicitation A1–A3 these are the central use case.

IO-2: Multiple regionally-relevant categories are omitted: (i) regulatory-equivalence STS as a distinct decision rule is absent — STS in the benchmark targets general semantic proximity [Q41]; (ii) no task category covers SmPC, CTD module, or EU-RMP NER as a genre — DrBenchmark acknowledges no large French biomedical benchmark existed prior [Q16] and web search confirms no French NLP benchmark with confirmed CTD module text exists [WEB-6]; (iii) IDMP-driven structured extraction from regulatory text is unaddressed [WEB-16, WEB-17].

IO-3: Several included categories are largely irrelevant to the deployment: pharmacy-diploma MCQA evaluates declarative knowledge retrieval rather than NER/STS (FRENCHMEDMCQA-D11, FRENCHMEDMCQA-D14); abstract-level multi-label specialty classification (MORFITT, DEFT2021-cls) does not test span-level extraction (MORFITT-D7, DEFT2021-D16); ICD-10 chapter classification (DiaMED) is orthogonal to regulatory entity extraction (DIAMED-D6, DIAMED-D13); negation/speculation scope NER (CAS, ESSAI) is at most indirectly relevant (CAS-D5, ESSAI-D12).

IO-4: Category gaps harming content validity: absence of regulatory-equivalence STS task category; absence of SmPC/CTD/EU-RMP genre-specific NER task category; absence of IDMP-structured extraction task. Web search confirms these are field-level gaps, not just benchmark gaps [WEB-11, WEB-16].

ID	Evidence
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Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q4	"These tasks encompass part-of-speech (POS) tagging, named-entity recognition (NER), classification, question-answering (QA), and semantic textual similarity (STS)." (p.1)
Q16	"To our knowledge, aside the multilingual benchmark BigBio (Fries et al., 2022) which includes only 4 corpora for French and is initially intended for generative text completion under zero-shot scenario, no large benchmark specialized in the French biomedical field exists." (p.2)
Q39	"FrenchMedMCQA (Labrak et al., 2022) is a Multiple-Choice Question-Answering (MCQA) dataset for biomedical domain. It contains 3,105 questions coming from real exams of the French medical specialization diploma in pharmacy, integrating single and multiple answers. The first task consists of automatically identifying the set of correct answers among the 5 proposed for a given question. The second task consists of identifying the number of answers (between 1 and 5) supposedly correct for a given question." (p.4)
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q81	"Table 8 summarizes the results obtained on average by the considered MLMs when aggregating the tasks into one of the five designated categories: POS, NER, MCQA, MCC (Multi-class classification), MLC (Multi-label classification), or STS tasks." (p.8)
CAS-D5	tokens: ['sans', 'antécédent', 'familial', 'de', 'maladie', 'colique']; ner_tags: [0, 1, 2, 2, 2, 2] — Negation scope spanning clinical history phrase; shows annotation granularity
DIAMED-D6	Monsieur Y. O. âgé de 19 ans a été hospitalisé le 03 février 2012 pour une paraplégie évoluant depuis un mois. — Classic clinical case narrative register; entirely distinct from regulatory document genre
MORFITT-D7	La mise en place par cathétérisme d'une valve aortique (TAVI)... Le registre France-TAVI... va inclure plus de 3 000 cas — Abstract-level classification only; no token-level entity annotations for NER task
FRENCHMED MCQA-D11	Au cours de la leucémie lymphoïde chronique, le myélogramme montre: — MCQA format — tests knowledge retrieval, not NER or STS; task type mismatch
ESSAI-D12	le traitement par un inhibiteur de PARP, le talazoparib, peut être plus efficace et mieux toléré de la chimiothérapie de référence — Speculation NER marks hedging scope, not INN entity
DIAMED-D13	Il s'agit d'un adolescent âgé de 16 ans, sportif (tennis) qui se plaignait depuis plusieurs mois de douleurs mécaniques du coude droit. — Clinical case narrative about orthopedic condition; no pharmaceutical content
FRENCHMED MCQA-D14	Parmi les propriétés suivantes du monoxyde de carbone, indiquer celle(s) qui est (sont) exacte(s) — Classification of true/false propositions — no STS pairwise similarity signal
DEFT2021-D16	features: id, document_id, text, specialities, specialities_one_hot — No STS pair structure; classification only; cannot evaluate regulatory equivalence scoring
WEB-6	DrBenchmark paper confirms task inventory of 20 tasks across the five categories; no SmPC/CTD/EU-RMP-specific task category present (https://arxiv.org/html/2402.13432v1)
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-16	IQVIA fact sheet — IDMP requires structured extraction from unstructured SmPC text; ~70% of required attributes reside in unstructured text; this task category is unaddressed by DrBenchmark (https://www.iqvia.com/-/media/iqvia/pdfs/library/fact-sheets/nlp-text-analytics-for-regulatory-affairs.pdf)
WEB-17	https://www.linguamatics.com/blog/countdown-idmp-compliance-are-you-ready

Information Gaps

- No documented evaluation category for regulatory-equivalence STS or IDMP-structured extraction; whether the benchmark's general STS performance is at all predictive of regulatory-equivalence performance is unknown.

Requires Expert Verification

- Whether deployment teams treat MANTRAGSC fr_patents as adequate proxy for pharmaceutical patent claim NER given its 20-example test set.

Remediation

Gap 1: No task category covers IDMP-driven structured extraction or regulatory-equivalence STS, both of which are central to the deployment [WEB-11, WEB-16].

Recommendation: Add two task categories to the evaluation suite: (a) IDMP attribute extraction from SmPC text targeting ATC, INN, excipient dose, therapeutic indication; (b) regulatory-equivalence STS over EMA-templated safety-warning pairs. Both can be designed as extensions of the existing HuggingFace toolkit [Q48].

ID	Evidence
Q48	"We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries." (p.5)
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-16	IQVIA fact sheet — IDMP requires structured extraction from unstructured SmPC text; ~70% of required attributes reside in unstructured text; this task category is unaddressed by DrBenchmark (https://www.iqvia.com/-/media/iqvia/pdfs/library/fact-sheets/nlp-text-analytics-for-regulatory-affairs.pdf)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input content overlaps only narrowly with the regulatory deployment. QUAERO's EMEA subset and Mantra-GSC's EMEA subset are the only two corpora containing genuine EMA-sourced text [\[Q33, Q34, Q40\]](#), and the dataset analysis confirms these are PIL-register drug leaflets rather than SmPC text, with QUAERO EMEA holding only ~10 segmented documents across train/dev/test (QUAERO-D1, QUAERO-D9, MANTRAGSC-D11, MANTRAGSC-D12) — corroborated by [\[WEB-3, WEB-4\]](#). The remaining majority of content is clinical case reports, research abstracts, clinical trial protocols, and spoken prescriptions [\[Q19, Q24, Q29, Q38, Q43, Q44, Q45\]](#), which differ substantially from EMA QRD-templated regulatory prose in register and formulaicity [\[WEB-2, WEB-7\]](#). While pharmaceutical vocabulary (INNs, dosages, routes) does appear across corpora (DEFT2021-D1, PXCORPUS-D1, ESSAI-D1, DEFT2020-D1), the deployment's target entity nomenclature (INN/ATC/exciipient distinctions, EMA-templated posology, contraindication qualifiers) is not realized in regulatory register at scale. Off-domain content (beekeeping, sports, history in DEFT2020; veterinary and non-EU geographies in MORFITT) further dilutes the relevant signal (DEFT2020-D13, DEFT2020-D14, MORFITT-D17, MORFITT-D18). Construct-irrelevant variance is therefore substantial, though not total.

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
Q24	"DEFT-2021 (Grouin et al., 2021) is a subset of 275 clinical cases taken from the 2019 edition of DEFT." (p.3)
Q29	"E3C (Magnini et al., 2020) is a multilingual dataset of clinical cases annotated for the NER task." (p.3)
Q33	"The QUAERO French Medical Corpus (Névéol et al., 2014), simply referred to as QUAERO in this paper, contains annotated entities and concepts for NER tasks." (p.3)
Q34	"The dataset covers two text genres (drug leaflets and biomedical titles), consisting of a total of 103,056 words sourced from EMEA or MEDLINE." (p.3)
Q38	"MorFITT (Labrak et al., 2023) is a multi-label dataset annotated with medical specialties. It contains 3,624 biomedical abstracts from PMC Open Access. It has been annotated across 12 medical specialties (more detail in Appendix B.5), for a total of 5,116 annotations." (p.4)
Q40	"Mantra-GSC (Kors et al., 2015) is a multilingual dataset annotated for biomedical NER. From the 5 languages covered, we included only the French subset in this benchmark. The dataset is obtained from 3 sources which have been partitioned to be evaluated separately by 2 annotation schemes (more detail in Appendix B.6): Medline (11 classes), and EMEA and Patents (10 classes). The sources cover different types of documents (biomedical abstracts/titles, drug labels and patents). To ensure evaluation consistency, we randomly split the dataset into 3 subsets: 70% for training, 10% for validation, and 20% for testing." (p.4)
Q43	"ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus." (p.4)
Q44	"PxCorpus (Kocabiyikoglu et al., 2022) is a spoken language understanding dataset in the domain of medical drug prescription transcripts. It includes 4 hours (1,981 recordings) of transcribed and annotated dialogues focused on drug prescriptions. The recordings were manually transcribed and semantically annotated. The first task involves classifying the textual utterances into one of the 4 intent classes (prescribe, replace, negate, none). The second task is a NER task where each word in a sequence is classified into one of 38 classes, such as drug, dose, or mode (more detail in Appendix B.9)." (p.4)

Q45	“DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session.” (p.4)
QUAERO-D1	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg/ml); phosphate de sodium, monobasique, monohydraté; chlorure de sodium; polysorbate 80 (E433) et eau pour préparation injectable. — Pharmaceutical composition list from SmPC-like regulatory text, annotating active substance and excipients
DEFT2021-D1	clarithromycine 500 mg deux fois par jour pendant 10 jours... sulfate de quinidine 200 mg trois fois par jour — Drug names with explicit dosage and frequency — relevant to deployment NER targets
PXCORPUS-D1	primperan 10 milligrammes comprimés 1 comprimé en cas de nausée toutes les 8 heures pendant 14 jours — Full prescription structure with drug name, dose, form, frequency, indication, duration
ESSAI-D1	avec la combinaison gemcitabine + abraxane, chez des patients avec un cancer du pancréas — Drug INN + brand name co-occurrence in trial text
DEFT2020-D1	En raison de la présence de lactose, ce médicament est contre-indiqué en cas de galactosémie congénitale, de syndrome de malabsorption du glucose et du galactose ou de déficit en lactase. — Contraindication phrasing with excipient (lactose) — directly deployment-relevant regulatory text
QUAERO-D9	La Commission européenne a délivré une autorisation de mise sur le marché valide dans toute l'Union européenne pour Tysabri à Elan Pharma International Ltd, le 27 juin 2006. — EMA marketing authorization reference in regulatory-genre French
MANTRAGSC-D11	train=70, validation=10, test=20 — Test set of 20 examples is statistically insufficient for robust NER evaluation
MANTRAGSC-D12	train=70, validation=10, test=20 — Same size constraint for patent genre evaluation
DEFT2020-D13	Certains apiculteurs sélectionnent leurs reines afin de favoriser au mieux la production. — Beekeeping encyclopedic content — irrelevant to regulatory deployment
DEFT2020-D14	Les Canadiens de Montréal sont une franchise de hockey sur glace professionnel située à Montréal dans la province de Québec au Canada. — Sports content — zero regulatory relevance
MORFITT-D17	L'otite externe est une maladie multifactorielle fréquente chez le chien... Décrire le microbiote bactérien auriculaire des chiens avec otite externe — Canine veterinary content irrelevant to human drug labeling
MORFITT-D18	Prévalence et caractéristiques morphologiques et moléculaires de Sarcocystis bertrami chez les chevaux en Chine — Equine parasitology in China; fully outside deployment scope
WEB-2	https://learning.eupati.eu/mod/book/view.php?id=858&chapterid=770
WEB-3	QUAERO EMEA HuggingFace dataset card confirms PIL leaflet text, not SmPC text (https://huggingface.co/datasets/DrBenchmark/QUAERO)
WEB-4	QUAERO official corpus page documents EMEA/MEDLINE source genres (https://quaerofrenchmed.limsi.fr/)
WEB-7	EMA QRD-templated documents have formulaic typographic conventions (section numbering, tabular posology) that may stress sentence-splitting at boundaries (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)

Strengths

- QUAERO EMEA contains genuine EMA drug leaflet text with composition lists, posology, contraindication blocks, and EMA administrative references (QUAERO-D1, QUAERO-D6, QUAERO-D9)
- DEFT2020 Task 1 includes drug leaflet sentence pairs with contraindication phrasing, dose-time thresholds, and excipient warnings (DEFT2020-D1, DEFT2020-D2, DEFT2020-D9)
- Pharmaceutical entity vocabulary (INN, dosage, route, frequency, duration) is densely present across multiple corpora — providing concept exposure even where regulatory-register annotation is absent (DEFT2021-D1, PXCORPUS-D1, ESSAI-D1)
- Patent claim genre is represented in MANTRAGSC fr_patents [Q40] (MANTRAGSC-D7, MANTRAGSC-D8)

ID	Evidence
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Q40	"Mantra-GSC (Kors et al., 2015) is a multilingual dataset annotated for biomedical NER. From the 5 languages covered, we included only the French subset in this benchmark. The dataset is obtained from 3 sources which have been partitioned to be evaluated separately by 2 annotation schemes (more detail in Appendix B.6): Medline (11 classes), and EMEA and Patents (10 classes). The sources cover different types of documents (biomedical abstracts/titles, drug labels and patents). To ensure evaluation consistency, we randomly split the dataset into 3 subsets: 70% for training, 10% for validation, and 20% for testing." (p.4)
QUAERO-D1	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg/ml); phosphate de sodium, monobasique, monohydraté; chlorure de sodium; polysorbate 80 (E433) et eau pour préparation injectable. — Pharmaceutical composition list from SmPC-like regulatory text, annotating active substance and excipients
DEFT2020-D1	En raison de la présence de lactose, ce médicament est contre-indiqué en cas de galactosémie congénitale, de syndrome de malabsorption du glucose et du galactose ou de déficit en lactase. — Contraindication phrasing with excipient (lactose) — directly deployment-relevant regulatory text
DEFT2021-D1	clarithromycine 500 mg deux fois par jour pendant 10 jours... sulfate de quinidine 200 mg trois fois par jour — Drug names with explicit dosage and frequency — relevant to deployment NER targets
PXCORPUS-D1	primperan 10 milligrammes comprimés 1 comprimé en cas de nausée toutes les 8 heures pendant 14 jours — Full prescription structure with drug name, dose, form, frequency, indication, duration
ESSAI-D1	avec la combinaison gemcitabine + abraxane, chez des patients avec un cancer du pancréas — Drug INN + brand name co-occurrence in trial text
DEFT2020-D2	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — Drug safety warning with dose-time threshold — deployment-central entity type
QUAERO-D6	N'utilisez jamais TYSABRI [...] Si vous êtes allergique (hydpersensible) au natalizumab ou à l'un des autres composants contenus dans TYSABRI [...] Si vous avez des perturbations graves du système immunitaire (dues à une maladie, telle que leucémie ou infection à VIH...) — Full contraindication block with population qualifiers, close to EU label format
MANTRAGSC-D7	Forme posologique orale selon la revendication 1, dans laquelle ledit antagoniste opioïde libérable est la naltrexone et ledit opioïde libérable est la codéine, dans laquelle le rapport de la naltrexone sur la codéine libérable est de 0,005 : 1 à 0,044 : 1. — French patent claim with INNs and dosage ratio; patent claim register confirmed
MANTRAGSC-D8	L' utilisation selon la revendication 3, où le laxatif osmotique est du glycol de polyéthylène 3350. — Excipient/drug substance in patent claim; CHEM tag does not distinguish excipient from INN
QUAERO-D9	La Commission européenne a délivré une autorisation de mise sur le marché valide dans toute l'Union européenne pour Tysabri à Elan Pharma International Ltd, le 27 juin 2006. — EMA marketing authorization reference in regulatory-genre French
DEFT2020-D9	- Ne pas utiliser chez les personnes présentant des difficultés de déglutition en raison du risque d'inhalation bronchique et de pneumopathie lipoïde. — Contraindication safety warning in EU leaflet formulaic register

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires region-specific knowledge: EMA QRD-templated phrasing (current v10.4, Feb 2024 [WEB-7, WEB-18]), EU directive 2001/83/EC compliance language [WEB-7], INN/ATC nomenclature, EMA administrative numbering (e.g., EMEA/H/C/...). QUAERO-D9 and QUAERO-D7 show occasional appearance of these conventions but they are not systematically represented.

IC-2: The benchmark content is in French biomedical register but predominantly clinical/research, not regulatory. EMA-templated standard statements ('est contre-indiqué chez les patients...', 'la posologie recommandée est de...') [WEB-7] appear only sporadically in QUAERO EMEA and DEFT2020 drug-leaflet pairs; the dominant register is narrative clinical prose

(CAS-D8, DIAMED-D6, E3C-D4).

IC-3: Western-specific knowledge per se transfers (deployment is EU-French), but several corpora contain content from non-EU contexts: MORFITT includes Canadian (MORFITT-D13, MORFITT-D14), Japanese (MORFITT-D15), and Saudi Arabian (MORFITT-D16) studies; DiaMED is sourced from The Pan African Medical Journal, with Sub-Saharan and North African settings (DIAMED-D9, DIAMED-D11). These do not transfer cleanly to EMA/ANSM jurisdictional context.

IC-4: INSUFFICIENT DOCUMENTATION — no regional regulatory annotators were recruited; benchmark author affiliations are academic NLP and clinical medicine [Q8], not regulatory affairs. Recruiting regulatory affairs specialists for QUAERO EMEA and MANTRAGSC fr_emea re-review would be the natural mitigation.

IC-5: Documented content issues harming content validity: (a) ~10 segmented PIL documents in QUAERO EMEA insufficient for SmPC-register evaluation [WEB-3, WEB-4]; (b) heterogeneous off-domain content in DEFT2020 (DEFT2020-D13, DEFT2020-D14, DEFT2020-D15) and MORFITT (MORFITT-D17, MORFITT-D18, MORFITT-D19) introduces construct-irrelevant variance; (c) PxCorpus spoken/ASR register (PXCORPUS-D10, PXCORPUS-D12) is incompatible with formal regulatory text; (d) QUAERO sentence-splitting artifacts (QUAERO-D19, QUAERO-D20) corrupt boundary-spanning regulatory phrasing.

ID	Evidence
Q8	"LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille" (p.1)
Q34	"The dataset covers two text genres (drug leaflets and biomedical titles), consisting of a total of 103,056 words sourced from EMEA or MEDLINE." (p.3)
DEFT2020-D1	En raison de la présence de lactose, ce médicament est contre-indiqué en cas de galactosémie congénitale, de syndrome de malabsorption du glucose et du galactose ou de déficit en lactase. — Contraindication phrasing with excipient (lactose) — directly deployment-relevant regulatory text
E3C-D4	La patiente a déjà mené une première grossesse à terme il y a 9 ans, donnant naissance à un petit garçon en bonne santé. — Narrative patient history; typical clinical case format, not regulatory document
DIAMED-D6	Monsieur Y. O. âgé de 19 ans a été hospitalisé le 03 février 2012 pour une paraplégie évoluant depuis un mois. — Classic clinical case narrative register; entirely distinct from regulatory document genre
QUAERO-D7	EMA/H/C/122 REFLUDAN... Son principe actif est la lépirudine. — Brand name and INN both tagged CHEM, demonstrates INN extraction capability
CAS-D8	Il s'agit d'une patiente âgée de 63 ans , sans antécédents particuliers , ménopausée depuis 14 ans , célibataire , sans enfants , qui a consulté pour algies pelviennes chroniques... — Prototypical clinical case narrative register, fundamentally different from regulatory document prose
QUAERO-D9	La Commission européenne a délivré une autorisation de mise sur le marché valide dans toute l'Union européenne pour Tysabri à Elan Pharma International Ltd, le 27 juin 2006. — EMA marketing authorization reference in regulatory-genre French
DIAMED-D9	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Sub-Saharan African hospital setting; non-EU regulatory context
PXCORPUS-D10	/chet — Single-token ASR artifact with no lexical content; not regulatory text
DIAMED-D11	suture de la rupture gastrique par un surjet au polyglactin 910 (vicryl*2/0) — Trade name usage in surgical narrative; clinical protocol differs from regulatory document conventions
MANTRAGSC-D11	train=70, validation=10, test=20 — Test set of 20 examples is statistically insufficient for robust NER evaluation
PXCORPUS-D12	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français — English/French code-switching and informal register unlike regulatory prose
MORFITT-D13	La région du Nord-Ouest de l'Ontario présente un taux élevé et documenté d'infections de la peau... causées par une souche de Staphylococcus aureus méthycillino-résistante d'origine communautaire — Canadian clinical context; jurisdictional mismatch with EMA/ANSM
DEFT2020-D13	Certains apiculteurs sélectionnent leurs reines afin de favoriser au mieux la production. — Beekeeping encyclopedic content — irrelevant to regulatory deployment
MORFITT-D14	En 2007 et 2008, 4 423 adultes de Calgary ont répondu à des entrevues au téléphone fixe — Canadian population study; non-EU context

DEFT2020-D14	Les Canadiens de Montréal sont une franchise de hockey sur glace professionnel située à Montréal dans la province de Québec au Canada. — Sports content — zero regulatory relevance
MORFITT-D15	Évaluer la charge des maladies d'origine alimentaire au Japon... pour l'année 2011 — Japanese epidemiological study; non-EU regulatory context
DEFT2020-D15	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605), gouverne la Russie à partir de 1594... — Historical encyclopedic content — off-domain for regulatory NLP
MORFITT-D16	La présente étude transversale a étudié les attitudes de médecins internes en Arabie saoudite — Saudi Arabian medical survey; non-EU jurisdiction
MORFITT-D17	L'otite externe est une maladie multifactorielle fréquente chez le chien... Décrire le microbiote bactérien auriculaire des chiens avec otite externe — Canine veterinary content irrelevant to human drug labeling
MORFITT-D18	Prévalence et caractéristiques morphologiques et moléculaires de Sarcocystis bertrami chez les chevaux en Chine — Equine parasitology in China; fully outside deployment scope
QUAERO-D19	Aucune étude clinique spécifique sur les interactions médicamenteuses n Toutefois, en raison des faibles concentrations plasmatiques du ziconotide... — Truncated mid-sentence artifact from document splitting
MORFITT-D19	Le nombre croissant de colonies d'abeilles mellifères et d'apiculteurs au Canada... lutter contre les maladies — Beekeeping disease management; unrelated to pharmaceutical regulatory deployment
QUAERO-D20	r. — Single-character fragment, sentence-splitting artifact
WEB-3	QUAERO EMEA HuggingFace dataset card confirms PIL leaflet text, not SmPC text (https://huggingface.co/datasets/DrBenchmark/QUAERO)
WEB-4	QUAERO official corpus page documents EMEA/MEDLINE source genres (https://quaerofrenchmed.limsi.fr/)
WEB-7	EMA QRD-templated documents have formulaic typographic conventions (section numbering, tabular posology) that may stress sentence-splitting at boundaries (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)
WEB-18	EMA QRD v10.4 template PDF — current operational template structure (https://www.ema.europa.eu/en/documents/template-form/qrd-product-information-template-version-104-highlighted_en.pdf)

Information Gaps

- No quantification of what proportion of QUAERO EMEA or DEFT2020 Task 1 content overlaps with current EMA QRD v10.4 template phrasing.
- No evidence on whether the benchmark's pharmaceutical vocabulary coverage transfers to current EMA SmPC vocabulary frequency distributions.

Requires Expert Verification

- Whether QUAERO EMEA's drug selection (Tysabri, Refludan, Prialt) is representative of current EMA centrally authorized product population.
- Whether MANTRAGSC fr_patents claim language matches contemporary pharmaceutical patent drafting conventions.

Remediation

Gap 1: No SmPC, CTD module, or EU-RMP text is present in any DrBenchmark dataset; QUAERO EMEA holds only ~10 segmented PIL documents [WEB-3, WEB-4].

Recommendation: Construct a supplementary corpus of French-translated EMA SmPCs (centralized procedure outputs are publicly available) covering at least the major ATC therapeutic classes, with stratified train/dev/test splits and regulatory-register-aware sentence segmentation that preserves section 4.x boundaries.

ID	Evidence
WEB-3	QUAERO EMEA HuggingFace dataset card confirms PIL leaflet text, not SmPC text (https://huggingface.co/datasets/DrBenchmark/QUAERO)
WEB-4	QUAERO official corpus page documents EMEA/MEDLINE source genres (https://quaerofrenchmed.limsi.fr/)

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is largely well-aligned. Both deployment and benchmark operate on French text in Latin script with standard diacritics, and the benchmark filters multilingual datasets to French-only [\[Q31\]](#). No script, modality, or infrastructure mismatch exists for the core deployment. Two concerns reduce the score from 5: (1) sentence-splitting of long EMEA documents [\[Q37\]](#) introduces structural artifacts (QUAERO-D19, QUAERO-D20) and removes document-level context that SmPC/CTD reasoning may require; and (2) PxCorpus introduces transcribed speech with ASR artifacts (PXCORPUS-D10, PXCORPUS-D12, PXCORPUS-D13) that are absent from the deployment's text-only document-management input modality. The benchmark also exposes non-French configs in E3C and MANTRAGSC HuggingFace repositories (E3C-D8, E3C-D9) that could contaminate evaluation if pipelines do not filter correctly. Tokenization variation is studied [\[Q78–Q80\]](#) and shown to be a minor effect.

ID	Evidence
Q31	"While the dataset covers 5 languages, only the French portion is retained for the benchmark." (p.3)
Q37	"Due to the presence of nested entities in annotations, we simplified the evaluation process by retaining only annotations at the higher granularity level from the BigBio (Fries et al., 2022) implementation, following the approach described in Touchent et al. (2023), which translates into an average loss of 6.06% of the annotations on EMEA and 8.90% on MEDLINE. Additionally, considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences." (p.3)
Q78	"However, our experiments, as shown in Table 7, reveal that FlauBERT is the model with the least word segmentation (1.43 in average), while DrBERT-CP tends to have the highest average segmentation (1.90 in average)." (p.8)
Q79	"Surprisingly, when comparing the performance of these two models on the benchmark tasks, we observe that DrBERT-CP outperforms FlauBERT on 16 out of the 20 tasks, thus contradicting previous conclusions drawn by the community." (p.8)
Q80	"Yet, tokenization, as it is currently done in MLMs, seems to play a minor role in the performance of systems." (p.8)
E3C-D8	Azken hilabeteetan lkernek kodeina + ibuprofenoa hartu ditu, baina ez du hobekuntza handirik nabaritu. — Basque text; completely outside French regulatory deployment scope
E3C-D9	A chest radiograph showed right upper lobe consolidation with volume loss, right para-tracheal and left hilar adenopathy — English clinical text; irrelevant to French deployment
PXCORPUS-D10	/chet — Single-token ASR artifact with no lexical content; not regulatory text
PXCORPUS-D12	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français — English/French code-switching and informal register unlike regulatory prose
PXCORPUS-D13	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment — System dialogue prompt embedded in training example; not a regulatory text pattern
QUAERO-D19	Aucune étude clinique spécifique sur les interactions médicamenteuses n Toutefois, en raison des faibles concentrations plasmatiques du ziconotide... — Truncated mid-sentence artifact from document splitting
QUAERO-D20	r. — Single-character fragment, sentence-splitting artifact

Strengths

- French-only filtering of multilingual corpora ensures language match [\[Q31\]](#)
- Latin script with standard French diacritics matches deployment writing system; no script-level NLP challenges

- HuggingFace toolkit with normalized loaders and predefined splits supports reproducible evaluation [Q48]
- Sentence-splitting addresses long-document input length constraints [Q37], a real challenge for SmPC and CTD documents

ID	Evidence
Q31	"While the dataset covers 5 languages, only the French portion is retained for the benchmark." (p.3)
Q37	"Due to the presence of nested entities in annotations, we simplified the evaluation process by retaining only annotations at the higher granularity level from the BigBio (Fries et al., 2022) implementation, following the approach described in Touchent et al. (2023), which translates into an average loss of 6.06% of the annotations on EMEA and 8.90% on MEDLINE. Additionally, considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences." (p.3)
Q48	"We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries." (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions broadly match: French Latin-script text in both contexts. Specific deviation: PxCorpus is transcribed speech with ASR noise (PXCORPUS-D9 'tropatepine chloridrate 1o milligrammes', PXCORPUS-D11 truncations) — distribution mismatch with deployment's structured PDF/Word document input.

IF-2: Yes — deployment is enterprise document management with standard text processing infrastructure; HuggingFace toolkit-based delivery [Q48] is compatible with on-premises or cloud pipelines.

IF-3: Domain-specific form differences: (a) EMEA documents sentence-split [Q37], potentially removing document-level structural context (section 4.1–4.9 hierarchy) that regulatory reasoning depends on; (b) DEFT2020 contains tabular data fragments mixed with prose (CLISTER-D19, CLISTER-D20) that would not occur in deployment's main prose extraction; (c) PxCorpus speech transcripts include code-switching and disfluencies (PXCORPUS-D11, PXCORPUS-D12) entirely absent from regulatory documents.

IF-4: Form mismatches harming external validity: sentence-splitting may hinder evaluation of long-document NLP needed for SmPC/CTD; spoken-language register in PxCorpus does not transfer to regulatory text. Both are partial concerns rather than fatal flaws since the bulk of NER/STS data is written-text French.

ID	Evidence
Q31	"While the dataset covers 5 languages, only the French portion is retained for the benchmark." (p.3)
Q37	"Due to the presence of nested entities in annotations, we simplified the evaluation process by retaining only annotations at the higher granularity level from the BigBio (Fries et al., 2022) implementation, following the approach described in Touchent et al. (2023), which translates into an average loss of 6.06% of the annotations on EMEA and 8.90% on MEDLINE. Additionally, considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences." (p.3)
Q48	"We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries." (p.5)
PXCORPUS-D9	tropatepine chloridrate 1o milligrammes 1 comprimé le midi — INN + salt form with minor transcription error ("1o" for "10"), showing ASR noise

PXCORPUS-D 10	/chet — Single-token ASR artifact with no lexical content; not regulatory text
PXCORPUS-D 11	ne pas tenir compte à midi tous les jours merd/ — Profanity and truncation in spoken transcript; absent from regulatory documents
PXCORPUS-D 12	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français — English/French code-switching and informal register unlike regulatory prose
QUAERO-D19	Aucune étude clinique spécifique sur les interactions médicamenteuses n Toutefois, en raison des faibles concentrations plasmatiques du ziconotide... — Truncated mid-sentence artifact from document splitting
CLISTER-D19	Jour 45 - - - - - 25 µg 0,4 mg" / "Jour 53 15,13 12,0 1,58 - - - - - 25 µg 0,4 mg — Table row fragments, not natural-language sentences
CLISTER-D20	Jour 91 - - - - - Fin 0,4 mg" / "Jour 187 1,67 21,2 1,38 - - - - - 0,4 mg — Numeric table rows without natural-language predication
QUAERO-D20	r. — Single-character fragment, sentence-splitting artifact

Information Gaps

- No quantification of how often sentence-splitting fragments cross regulatory phrase boundaries (e.g., dose-threshold qualifiers split from drug name).

Remediation

Gap 1: Sentence-splitting of long EMEA documents [Q37] removes document-level structural context (QUAERO-D19, QUAERO-D20) that SmPC/CTD reasoning depends on.

Recommendation: Augment the QUAERO EMEA loader with a section-aware splitter that preserves SmPC section identifiers (4.1–4.9) as input metadata, and evaluate models with and without this structural context to quantify document-level reasoning effects.

ID	Evidence
Q37	“Due to the presence of nested entities in annotations, we simplified the evaluation process by retaining only annotations at the higher granularity level from the BigBio (Fries et al., 2022) implementation, following the approach described in Touchent et al. (2023), which translates into an average loss of 6.06% of the annotations on EMEA and 8.90% on MEDLINE. Additionally, considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences.” (p.3)
QUAERO-D19	Aucune étude clinique spécifique sur les interactions médicamenteuses n Toutefois, en raison des faibles concentrations plasmatiques du ziconotide... — Truncated mid-sentence artifact from document splitting
QUAERO-D20	r. — Single-character fragment, sentence-splitting artifact

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output ontology shows fundamental misalignment with the deployment, which the elicitation marked HIGH priority. Two critical violations are present. (1) STS scoring uses general semantic proximity on a 0–5 scale [Q41, Q54], whereas the deployment requires regulatory-equivalence scoring sensitive to dose-threshold and population-qualifier micro-differences (elicitation A3). The dataset analysis empirically demonstrates the calibration is incompatible: a 2× quantitative difference scores 4.0, a 14× difference scores 3.75, and a 60× PSA difference scores 3.0 (CLISTER-D12, CLISTER-D14, CLISTER-D13); legally significant additions of specific opioid names or 'sous stricte surveillance médicale' qualifiers are not penalized in DEFT2020 (DEFT2020-D16, DEFT2020-D17, DEFT2020-D20). (2) NER label categories are calibrated to clinical/UMLS taxonomies that conflate distinct regulatory entity types: QUAERO and Mantra-GSC's CHEM tag merges INN, brand name, and excipient (QUAERO-D10, QUAERO-D11, MANTRAGSC-D5, MANTRAGSC-D14); ATC codes and EMA administrative identifiers receive O tag (QUAERO-D12); contraindication qualifiers are tagged by entity type rather than as regulatory contraindication structure (QUAERO-D16, MANTRAGSC-D4). Web search confirms no French NLP benchmark with INN/ATC/excipient/contraindication-specific schemas exists [WEB-11, WEB-12]. The leading model also does not excel at MLC or STS [Q82] — the dimensions on which the deployment most depends. This is a structural validity violation.

ID	Evidence
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q54	"For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020)." (p.6)
Q82	"Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS." (p.8)
MANTRAGSC-D4	Vous ne devez pas prendre Pramipexole Teva si vous allaitez. — Contraindication in patient leaflet register; condition tagged DISO not as contraindication type
MANTRAGSC-D5	1 ml de solution contient 40 microgrammes de travoprost et 5 mg de timolol (sous forme de maléate de timolol) — INN and salt excipient form both tagged CHEM; no INN/excipient distinction
QUAERO-D10	Contient également: mannitol, hydroxyde de sodium. — Excipients tagged identically to INNs (CHEM), preventing entity sub-class distinction
QUAERO-D11	Refludan 50 mg en poudre pour solution injectable... le mannitol (E 421) et l'hydroxyde de sodium pour l'ajustement du pH. — Excipient same tag as INN, conflating distinct regulatory entity types
CLISTER-D12	L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — 2× quantitative difference scored 4.0 (very similar)
QUAERO-D12	EMA/H/C/122 — EMA product number tagged O (no entity), missing key regulatory identifier class
CLISTER-D13	Le taux de PSA était de 218 ng/ml (normale ≤ 4ng/ml)." / "Le dosage de PSA était de 13327 ng/ml (normal : < 4 ng/ml). — 60× magnitude difference scored 3.0
CLISTER-D14	Le taux de PSA était de 4,49 ng/ml." / "Le taux de PSA total était à 66,68 ng/ml. — 14× numeric discrepancy scored near-similar
MANTRAGSC-D14	Renagel 800 mg sevelamer — Brand name and INN both CHEM; no proprietary/INN label distinction critical for regulatory workflows

DEFT2020-D16	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High similarity score despite addition of specific drug names in target — legally significant detail unmarked
QUAERO-D16	Par conséquent, Refludan ne doit pas être administré à la femme enceinte ou qui allaite. — Contraindication clause with pregnancy qualifier — UMLS tags entity type but not regulatory contraindication structure
DEFT2020-D17	Chez les patients insuffisamment équilibrés par glimepiride arrow à la dose maximale, un traitement par l'insuline peut être associé si nécessaire. — "sous stricte surveillance médicale" absent in source; critical qualifier not flagged in score
DEFT2020-D20	L'absorption de boissons alcoolisées est fortement déconseillée pendant le traitement (potentialisation réciproque). — Mechanism qualifier "(potentialisation réciproque)" dropped in cible; not flagged as regulatory deviation by score
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-12	Bayer et al. 2021 — used pharmacovigilance evaluators for ADE extraction but not regulatory equivalence scoring; closest comparator confirms gap (https://link.springer.com/article/10.1007/s40264-020-00996-3)

Strengths

- Pharmaceutical entity types are partially captured: PxCorpus has 38 NER classes including DRUG, DOSE, MODE, FREQUENCY, DURATION [Q44, Q108] (PXCORPUS-D1, PXCORPUS-D2)
- DEFT2021 NER includes SUBSTANCE, DOSAGE, MODE, DURATION as deployment-relevant entity classes (DEFT2021-D3, DEFT2021-D4)
- Replace intent class in PxCorpus directly captures dosage correction acts analogous to compliance flagging (PXCORPUS-D6, PXCORPUS-D7)
- STS exists as an output type with continuous 0–5 scoring and Spearman/EDRM evaluation [Q54], even if calibration mismatches the deployment

ID	Evidence
Q44	"PxCorpus (Kocabiyikoglu et al., 2022) is a spoken language understanding dataset in the domain of medical drug prescription transcripts. It includes 4 hours (1,981 recordings) of transcribed and annotated dialogues focused on drug prescriptions. The recordings were manually transcribed and semantically annotated. The first task involves classifying the textual utterances into one of the 4 intent classes (prescribe, replace, negate, none). The second task is a NER task where each word in a sequence is classified into one of 38 classes, such as drug, dose, or mode (more detail in Appendix B.9)." (p.4)
Q54	"For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020)." (p.6)
Q108	"Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE" (p.14)
PXCORPUS-D 1	primperan 10 milligrammes comprimés 1 comprimé en cas de nausée toutes les 8 heures pendant 14 jours — Full prescription structure with drug name, dose, form, frequency, indication, duration
PXCORPUS-D 2	ténofovir 245 milligrammes en comprimés à prendre après les repas 1 comprimé le soir pendant 2 semaines puis stop — INN drug name with dosage, route instruction, and duration
DEFT2021-D3	rifampicine – isoniazide – pyrazinamide pendant 6 mois — Drug names tagged as SUBSTANCE with duration — partial INN coverage
DEFT2021-D4	Hydrocortisone 300 mg IV... Hydroxyzine 25 mg par voie orale toutes les 6 heures — Drug + dose + route + frequency tagged; matches deployment posology targets
PXCORPUS-D 6	attention il s'agit de 20 milligrammes et pas 10 milligrammes — Explicit dosage correction; directly relevant to dose-discrepancy detection
PXCORPUS-D 7	remplacer 1 comprimé tous les jours par 1 comprimé en cas d'anxiété — Frequency/indication substitution captured by replace intent class

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance

OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories reviewed: NER schemas span UMLS Semantic Groups (10–11 classes in QUAERO/Mantra-GSC) [Q35, Q103, Q106], clinical entity layers (CLINENTITY/EVENT/ACTOR/etc. in E3C) [Q104], 38-class pharmaceutical schema (PxCorpus) [Q108], 13-type fine-grained NER (DEFT2021) [Q28], and classification schemas (MeSH-C 23 axes [Q27], 12 specialties [Q105], 22 ICD-10 chapters [Q45]). STS uses 0–5 continuous scale [Q41].

OO-2: Missing categories specific to the regional regulatory context: (a) INN as distinct from brand name and excipient — absent across all NER schemas (QUAERO-D7, QUAERO-D10, MANTRAGSC-D5); (b) ATC code as entity class — absent (QUAERO-D12), and confirmed by [WEB-11] as field-level gap; (c) excipient name as distinct regulatory entity — absent (QUAERO-D11, MANTRAGSC-D8); (d) EMA-templated posology structure — absent at template-fidelity level; (e) contraindication qualifier as regulatory construct — absent (QUAERO-D16, MANTRAGSC-D4); (f) regulatory-equivalence STS scale — absent; CLISTER calibration is incompatible (CLISTER-D12, CLISTER-D13, CLISTER-D14).

OO-3: Categories encoding non-regional or non-deployment values: ICD-10 chapter classification (DiaMED) reflects clinical disease taxonomy not regulatory document structure; MeSH-C axes (DEFT2021) reflect biomedical literature indexing not regulatory data attributes; UMLS Semantic Groups [Q35] are research-oriented and conflate regulatory-distinct chemicals into single CHEM class.

OO-4: Stakeholder-driven taxonomy redesign is required: regulatory affairs specialists would need to define a regulatory entity schema (INN/ATC/excipient/posology/contraindication-qualifier) and a regulatory-equivalence STS rubric, neither of which currently exists in DrBenchmark or, per web search, in any French NLP benchmark [WEB-11, WEB-12].

OO-5: Taxonomy issues harming structural and content validity: STS scoring rule fundamentally mismatches the deployment's regulatory-equivalence decision rule (CLISTER-D12, CLISTER-D13, CLISTER-D14, DEFT2020-D16, DEFT2020-D17, DEFT2020-D20); NER label space conflates regulatorily distinct entity types (QUAERO-D10, QUAERO-D11, MANTRAGSC-D14); regulatory-specific entity types are absent from all schemas (QUAERO-D12).

ID	Evidence
Q27	"The dataset is annotated with 23 axes from Chapter C of the Medical Subject Headings (MeSH)." (p.3)
Q28	"The second task involves fine-grained information extraction for 13 types of entities (more detail in Appendix B.7)." (p.3)
Q35	"10 entity categories corresponding to the UMLS Semantic Groups (Lindberg et al., 1993) were annotated (more detail in Appendix B.3)." (p.3)
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q54	"For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020)." (p.6)
Q82	"Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS." (p.8)
Q103	"O, GEOG, PHEN, DISO, ANAT, OBJC, PHYS, PROC, DEVI, CHEM and LIVB" (p.14)
Q104	"Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML" (p.14)

Q105	"microbiology, etiology, virology, physiology, immunology, parasitology, genetics, chemistry, veterinary, surgery, pharmacology and psychology" (p.14)
Q106	"Medline: ANAT, PROC, CHEM, PHYS, GEOG, DEVI, LIVB, OBJC, DISO, PHEN and O. EMEA and Patents: ANAT, PROC, CHEM, PHYS, DEVI, LIVB, OBJC, DISO, PHEN and O." (p.14)
Q108	"Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE" (p.14)
MANTRAGSC-D4	Vous ne devez pas prendre Pramipexole Teva si vous allaitez. — Contraindication in patient leaflet register; condition tagged DISO not as contraindication type
MANTRAGSC-D5	1 ml de solution contient 40 microgrammes de travoprost et 5 mg de timolol (sous forme de maléate de timolol) — INN and salt excipient form both tagged CHEM; no INN/excipient distinction
QUAERO-D7	EMEA/H/C/122 REFLUDAN... Son principe actif est la lépirudine. — Brand name and INN both tagged CHEM, demonstrates INN extraction capability
MANTRAGSC-D8	L' utilisation selon la revendication 3, où le laxatif osmotique est du glycol de polyéthylène 3350. — Excipient/drug substance in patent claim; CHEM tag does not distinguish excipient from INN
QUAERO-D10	Contient également: mannitol, hydroxyde de sodium. — Excipients tagged identically to INNs (CHEM), preventing entity sub-class distinction
QUAERO-D11	Refludan 50 mg en poudre pour solution injectable... le mannitol (E 421) et l'hydroxyde de sodium pour l'ajustement du pH. — Excipient same tag as INN, conflating distinct regulatory entity types
QUAERO-D12	EMEA/H/C/122 — EMA product number tagged O (no entity), missing key regulatory identifier class
CLISTER-D12	L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — 2× quantitative difference scored 4.0 (very similar)
CLISTER-D13	Le taux de PSA était de 218 ng/ml (normale ≤ 4ng/ml)." / "Le dosage de PSA était de 13327 ng/ml (normal : < 4 ng/ml). — 60× magnitude difference scored 3.0
CLISTER-D14	Le taux de PSA était de 4,49 ng/ml." / "Le taux de PSA total était à 66,68 ng/ml. — 14× numeric discrepancy scored near-similar
MANTRAGSC-D14	Renagel 800 mg sevelamer — Brand name and INN both CHEM; no proprietary/INN label distinction critical for regulatory workflows
QUAERO-D16	Par conséquent, Refludan ne doit pas être administré à la femme enceinte ou qui allaite. — Contraindication clause with pregnancy qualifier — UMLS tags entity type but not regulatory contraindication structure
DEFT2020-D16	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High similarity score despite addition of specific drug names in target — legally significant detail unmarked
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WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-12	Bayer et al. 2021 — used pharmacovigilance evaluators for ADE extraction but not regulatory equivalence scoring; closest comparator confirms gap (https://link.springer.com/article/10.1007/s40264-020-00996-3)

Information Gaps

- No quantification of how often clinical and regulatory NER label decisions would diverge on the same span (e.g., would CHEM-tagged 'natalizumab' match an INN-class regulatory annotation in a controlled comparison).

Requires Expert Verification

- Whether DEFT2021's SUBSTANCE/DOSAGE/MODE/DURATION schema, when applied to actual SmPC text, would yield acceptable regulatory-grade extraction or would require relabeling.
- Whether the EDRM metric [Q54] could be re-purposed for regulatory-equivalence scoring with a new annotation rubric.

Remediation

Gap 1: STS scoring is calibrated for general semantic proximity and demonstrably tolerates regulatorily significant differences (CLISTER 60× quantitative divergence scores 3.0; DEFT2020 specific drug-name additions score 4.0).

Recommendation: Develop a complementary regulatory-equivalence STS rubric with explicit penalties for dose-threshold changes, population-qualifier changes, and INN substitutions. Annotate a held-out evaluation set of EMA SmPC safety-warning sentence pairs against this rubric and report Spearman correlation separately from clinical CLISTER/DEFT2020 STS.

Gap 2: NER schemas conflate INN, brand name, and excipient under a single CHEM tag and do not annotate ATC codes or EMA administrative identifiers (QUAERO-D10, QUAERO-D12, MANTRAGSC-D14).

Recommendation: Define a regulatory NER schema with distinct INN, brand-name, excipient, ATC-code, EMA-identifier, and contraindication-qualifier classes. Re-annotate QUAERO EMEA and Mantra-GSC fr_emea (the only EMA-sourced subsets) using this schema, with regulatory affairs specialists as annotators.

ID	Evidence
QUAERO-D10	Contient également: mannitol, hydroxyde de sodium. — Excipients tagged identically to INNs (CHEM), preventing entity sub-class distinction
QUAERO-D12	EMA/H/C/122 — EMA product number tagged O (no entity), missing key regulatory identifier class
MANTRAGSC-D14	Renagel 800 mg sevelamer — Brand name and INN both CHEM; no proprietary/INN label distinction critical for regulatory workflows

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotator alignment with regulatory standards is largely undocumented and where documented, is inconsistent with the deployment's ground-truth norms. Annotation provenance is documented for only three datasets — DEFT-2021 (manual NER and MLC) **[Q25]**, CLISTER (multiple annotators averaged) **[Q41]**, DiaMED (several annotators including one medical expert, with Cohen's Kappa and Gwet's AC1) **[Q45, Q46]**, and CAS POS (automatic via Tagex 3) **[Q42]**. None of these involve regulatory affairs or pharmacovigilance specialists; benchmark author affiliations are academic NLP and clinical medicine **[Q8]**. The elicitation (A4) confirms systematic disagreement is anticipated on borderline cases. Empirical dataset evidence corroborates: CLISTER scoring patterns are inconsistent with regulatory-equivalence norms (CLISTER-D12, CLISTER-D17, CLISTER-D18); DEFT2020 annotators score legally significant qualifier additions/deletions as near-equivalent (DEFT2020-D16, DEFT2020-D17, DEFT2020-D20); QUAERO/Mantra-GSC clinical UMLS conventions tag pregnancy contraindications by entity type rather than regulatory construct (QUAERO-D16, MANTRAGSC-D4). Convergent and external validity are both threatened. The score is 2 rather than 1 because some annotation processes are documented with multi-annotator IAA (DiaMED) and CLISTER's exact-paraphrase ceiling behavior shows annotator reliability on unambiguous cases (DEFT2020-D8).

ID	Evidence
Q8	"LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille" (p.1)
Q25	"This dataset is manually annotated in two tasks: (i) multi-label classification and (ii) NER." (p.3)
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q46	"Table 4: Inter-annotator agreement statistics. κ is referring to Kappa Cohen and G to Gwet's AC1." (p.4)
MANTRAGSC-D4	Vous ne devez pas prendre Pramipexole Teva si vous allaitez. — Contraindication in patient leaflet register; condition tagged DISO not as contraindication type
DEFT2020-D8	Troubles de l'hémostase à type de maladie de Willebrand (se traduisant par un allongement du TCA, du temps de saignement et une diminution des taux du complexe VIIIc/VWF). — Identical pair, unanimous perfect score — confirms annotator reliability on exact match
CLISTER-D12	L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — 2× quantitative difference scored 4.0 (very similar)

DEFT2020-D16	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High similarity score despite addition of specific drug names in target — legally significant detail unmarked
QUAERO-D16	Par conséquent, Refludan ne doit pas être administré à la femme enceinte ou qui allaite. — Contraindication clause with pregnancy qualifier — UMLS tags entity type but not regulatory contraindication structure
CLISTER-D17	Métoprolol 50 mg deux fois par jour;" / "Metformine 500 mg, 1 comprimé deux fois par jour — Different INNs scored 2.0 due to structural template; regulatory judgment would score near 0
DEFT2020-D17	Chez les patients insuffisamment équilibrés par glimepiride arrow à la dose maximale, un traitement par l'insuline peut être associé si nécessaire. — "sous stricte surveillance médicale" absent in source; critical qualifier not flagged in score
CLISTER-D18	Ésoméprazole 20 mg/jour Hépatique T1/2 = 1,0 – 1,5 h / "Périndopril 2 mg/jour NR T1/2 = 17 h — Different INNs and PK profiles scored 2.0; inconsistent with regulatory equivalence
DEFT2020-D20	L'absorption de boissons alcoolisées est fortement déconseillée pendant le traitement (potentialisation réciproque). — Mechanism qualifier "(potentialisation réciproque)" dropped in cible; not flagged as regulatory deviation by score

Strengths

- DiaMED documents inter-annotator agreement using both Cohen's Kappa and Gwet's AC1 across two sessions [Q45, Q46]
- CLISTER uses multiple annotators with score averaging, providing variance signal that could be analyzed for borderline cases [Q41] (DEFT2020-D7)
- CAS POS annotation was validated against manual annotations at 98% precision [Q42], establishing baseline annotation quality for that subtask
- DiaMED includes one medical expert annotator [Q45]

ID	Evidence
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
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Q46	"Table 4: Inter-annotator agreement statistics. κ is referring to Kappa Cohen and G to Gwet's AC1." (p.4)
DEFT2020-D7	Après inspection, elles ont toutes été exclues. — High inter-annotator variance on borderline pair — illustrates scoring uncertainty

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels do not reflect regulatory affairs stakeholder perspectives. No annotation guidelines reference EMA/ANSM regulatory standards in the benchmark paper. Author affiliations [Q8] are academic NLP and clinical medicine institutions. The one documented medical expert (DiaMED) annotated ICD-10 classification, not the NER or STS tasks the deployment relies on [Q45].

OC-2: Systematic disagreement is expected per elicitation A4. Empirical dataset evidence shows specific divergence patterns: STS scoring tolerates $2\times\text{--}60\times$ quantitative differences (CLISTER-D12, CLISTER-D13, CLISTER-D14); legally significant qualifier additions/deletions are not penalized (DEFT2020-D16, DEFT2020-D17, DEFT2020-D20); different INNs receive non-zero similarity scores due to structural template matching (CLISTER-D17, CLISTER-D18); clinical UMLS conventions tag contraindication populations by entity type rather than as regulatory contraindication structure (QUAERO-D16, MANTRAGSC-D4).

OC-3: Annotator demographics are NOT DOCUMENTED for the majority of corpora. The benchmark paper provides no Datasheet- or Data-Statement-level demographic breakdown. Documented information is limited to: CLISTER 'multiple annotators' [Q41], DiaMED 'several annotators including one medical expert' [Q45], DEFT-2021 'manually annotated' [Q25], CAS automatic Tagex 3 with manual validation [Q42]. Professional backgrounds beyond 'medical expert' (DiaMED) are not characterized.

OC-4: Re-annotation by a representative regional pool (regulatory affairs specialists, EMA/ANSM-trained pharmacologists, regulatory legal experts) would be required to align ground truth with the deployment's interpretive norms. This is consistent with elicitation A4's expectation of systematic disagreement on borderline cases.

OC-5: Aggregation methods: CLISTER averages annotator scores into a floating-point reference [Q41], which can erase minority/borderline disagreement. DEFT2020-D7 illustrates high inter-annotator variance (scores [2.0, 4.5, 0.0, 1.0, 0.0]) collapsed to mean 1.5, hiding the disagreement that would be diagnostic for regulatory borderline cases.

OC-6: Label issues harming convergent and external validity: STS labels do not converge with regulatory-equivalence judgments (CLISTER-D12–14, DEFT2020-D16–D20); NER labels do not converge with regulatory-entity granularity (QUAERO-D10–D12, MANTRAGSC-D5, MANTRAGSC-D14); aggregation by averaging hides borderline disagreements that are precisely the deployment's high-stakes flag cases.

ID	Evidence
Q8	"LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille" (p.1)
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CLISTER-D14	Le taux de PSA était de 4,49 ng/ml." / "Le taux de PSA total était à 66,68 ng/ml. — 14× numeric discrepancy scored near-similar
MANTRAGSC-D14	Renagel 800 mg sevelamer — Brand name and INN both CHEM; no proprietary/INN label distinction critical for regulatory workflows
DEFT2020-D16	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High similarity score despite addition of specific drug names in target — legally significant detail unmarked
QUAERO-D16	Par conséquent, Refludan ne doit pas être administré à la femme enceinte ou qui allaite. — Contraindication clause with pregnancy qualifier — UMLS tags entity type but not regulatory contraindication structure
DEFT2020-D17	Chez les patients insuffisamment équilibrés par glimepiride arrow à la dose maximale, un traitement par l'insuline peut être associé si nécessaire. — "sous stricte surveillance médicale" absent in source; critical qualifier not flagged in score
CLISTER-D17	Métoprolol 50 mg deux fois par jour;" / "Metformine 500 mg, 1 comprimé deux fois par jour — Different INNs scored 2.0 due to structural template; regulatory judgment would score near 0
CLISTER-D18	Ésoméprazole 20 mg/jour Hépatique T1/2 = 1,0 – 1,5 h" / "Périndopril 2 mg/jour NR T1/2 = 17 h — Different INNs and PK profiles scored 2.0; inconsistent with regulatory equivalence
DEFT2020-D20	L'absorption de boissons alcoolisées est fortement déconseillée pendant le traitement (potentialisation réciproque). — Mechanism qualifier "(potentialisation réciproque)" dropped in cible; not flagged as regulatory deviation by score
WEB-11	Kadi et al. 2025 — confirms field-level gap; no regulatory-grade annotation rubric for SmPC extraction is established (https://pmc.ncbi.nlm.nih.gov/articles/PMC12380897/)
WEB-12	Bayer et al. 2021 — used pharmacovigilance evaluators for ADE extraction but not regulatory equivalence scoring; closest comparator confirms gap (https://link.springer.com/article/10.1007/s40264-020-00996-3)

Information Gaps

- No demographic breakdown of annotator pools across the majority of corpora.
- No published comparison study of biomedical NLP annotators vs. regulatory affairs specialists on the same regulatory text — confirmed field-level gap [annotator_alignment_verification, NOT FOUND].

Requires Expert Verification

- Magnitude of disagreement between benchmark labels and regulatory affairs judgments on a sampled set of QUAERO EMEA / Mantra-GSC fr_emea spans.
- Whether DiaMED's IAA methodology could serve as a template for re-annotating regulatory-relevant subsets with regulatory affairs annotators.

Remediation

Gap 1: Annotator demographics and professional backgrounds are undocumented for most corpora; no regulatory affairs annotators are documented anywhere in the benchmark [Q8, Q41, Q45].

Recommendation: Conduct a calibration study: have a panel of regulatory affairs specialists re-score a stratified sample of CLISTER, DEFT2020 drug-leaflet pairs, and QUAERO EMEA spans against benchmark labels. Quantify systematic disagreement and use the magnitude to weight or correct benchmark performance estimates.

ID	Evidence
Q8	“LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille” (p.1)

Q41	“CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided.” (p.4)
Q45	“DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session.” (p.4)

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is well-aligned. The deployment consumes label and score outputs in modalities the benchmark produces: IOB2 sequence labels via SeqEval [Q52], multiple-choice selections, multi-label classification labels, and continuous similarity scores [Q54]. Models predict only the first token label per word for tokenizer-agnostic evaluation [Q53]. Hyperparameters are documented [Q49, Q99] and best models saved on validation metrics [Q100]; results averaged over four runs with Student's t-test [Q51]. No output representation mismatch exists. Two minor concerns: (1) the EDRM metric [Q54] is calibrated for DEFT-2020 task design and has no documented regulatory calibration — this is a calibration concern that cuts across to OO; (2) compute requirements (~2,500 GPU hours [Q93], 36.9 kgCO2eq [Q94]) raise reproducibility concerns for resource-constrained users [Q92], though HuggingFace toolkit and documented hyperparameters [Q49] mitigate this. The output-form modality match is strong, justifying score 4 rather than 5.

ID	Evidence
Q49	"For further guidance, we have integrated all the training details, including hyperparameters, in Appendix A. This information will help users to reproduce and customize the experiments conducted with DrBenchmark." (p.5)
Q51	"All the models are fine-tuned regarding a strict protocol using the same hyperparameters for each downstream task. The reported results are obtained by averaging the scores from four separate runs, thus ensuring robustness and reliability. We also report statistical significance computed using Student's t-test." (p.5)
Q52	"To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023)." (p.6)
Q53	"It provides a tokenizer-agnostic evaluation and mitigates any correlation between models' performances and the tokenization process." (p.6)
Q54	"For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020)." (p.6)
Q92	"Although the benchmark is easily reproducible and customizable, it required a substantial amount of computational power to execute all runs." (p.9)
Q93	"We utilized approximately 2,500 hours on V100 GPUs from the Jean-Zay supercomputer to complete the quantitative study." (p.9)
Q94	"According to the Jean Zay supercomputer documentation, the total environmental cost is estimated to be equivalent to 647,500 Wh or 36.9 kgCO2eq/kWh, based on the carbon intensity of the energy grid mentioned in the BLOOM environmental cost study conducted on Jean Zay (Luccioni et al., 2022)." (p.9)
Q99	"For the experiments, we utilize the following hyperparameters that yield optimal performance from the models." (p.13)
Q100	"To mitigate overfitting, we locally save the best model based on its validation metric." (p.13)

Strengths

- IOB2 sequence labels via SeqEval are a deployment-compatible NER output format [Q52]
- Tokenizer-agnostic evaluation by predicting first-token label per word [Q53]
- Continuous STS scores with Spearman correlation provide compatible output modality even if scoring rule mismatches [Q54]
- Reproducibility supported by documented hyperparameters [Q49, Q99] and best-model saving on validation metric [Q100]

- Statistical significance reported via Student's t-test over four runs [Q51]

ID	Evidence
Q49	"For further guidance, we have integrated all the training details, including hyperparameters, in Appendix A. This information will help users to reproduce and customize the experiments conducted with DrBenchmark." (p.5)
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Q100	"To mitigate overfitting, we locally save the best model based on its validation metric." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality matches: deployment consumes label outputs (NER spans, classification decisions) and continuous scores (STS for safety-warning equivalence). Benchmark produces both [Q52, Q54]. No modality mismatch.

OF-2: Not applicable — deployment is text-only document management with no speech-output requirement [infrastructure_notes interface_modality].

OF-3: Target population is regulatory affairs specialists, pharmacologists, and legal experts — a professional cohort with full literacy. No accessibility requirement diverges from general French professional norms [target_population description].

OF-4: Form mismatches harming external validity: minimal at the representation level. The substantive concern (EDRM and Spearman correlation are not calibrated for regulatory equivalence detection) is properly attributed to OO (scoring rule), not OF (representation).

ID	Evidence
Q52	"To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023)." (p.6)
Q54	"For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020)." (p.6)